Language of logic

1

a $\sim$ **b** $\wedge$ **c** $\vee$ **d** $\Longrightarrow$

e $\Longleftrightarrow$

2

a $\sim p$ **b** $p \wedge q$ **c** $\sim q \vee \sim p$ **d** $\sim q \Longleftrightarrow \sim p$

e $\sim p \Longrightarrow \sim q$ **f** $\sim q \wedge p$

3

a $\sim q \Longleftrightarrow p$ **b** $p \Longrightarrow \sim q$ **c** $\sim p \wedge q$ **d** $\sim p \wedge \sim q$

e $\sim (p \vee q)$ **f** $\sim (\sim q \Longrightarrow p)$

4

a James does not have a smartphone.

b Dave has a laptop and James has a smartphone.

c If Dave does not have a laptop, then James has a smartphone.

d Dave does not have a laptop or James does not have a smartphone.

e It is false that Dave has a laptop and James has a smartphone.

5

a Quentin does not have a rabbit.

b Quentin has a rabbit or Maximilian has a dog.

c Quentin does not have a rabbit if and only if Maximilian does not have a dog.

d It is false that Quentin has a rabbit or Maximilian has a dog.

e Quentin does not have a rabbit and Maximilian does not have a dog.

6

a $(p \wedge \sim q) \wedge r$ **b** $(\sim p \wedge q) \wedge r$ **c** $(p \wedge q) \vee r$ **d** $p \Longrightarrow (q \vee \sim r)$

7

a $(r \wedge q) \Longrightarrow p$ **b** $(\sim p \Longleftrightarrow \sim q) \vee \sim r$

c $(r \Longleftrightarrow q) \wedge p$ **d** $q \Longrightarrow (p \Longleftrightarrow r)$

8

a The water is $82^{\circ}\mathrm{F}$ or the sun is down, and we do not go swimming.

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b The water is not 28°C, and the sun is down or we go swimming.

c If we do not go swimming, then the sun is down and the water is 28°C.

d If the sun is down then we go swimming, and the water is 28°C.

e The sun is down if and only if the water is 28°C, and we go swimming.

9

a The water is 60°F and the sun is up, or we go snorkelling.

b If the sun is up then the water is 60°F, or we go snorkelling.

c If the sun is up and we go snorkelling, then the water is 60°F.

d If the water is not 60°F, then the sun is up or we go snorkelling.

e If the sun is up, then the water is 60°F if and only if we go snorkelling.

10

a $p \wedge \sim q$ **b** Conjunction

11

a $p \implies (q \vee r)$ **b** Conditional

12

a $\sim (p \implies \sim q)$ **b** Negation

13

a $(p \implies q) \vee \sim r$ **b** Disjunction

14

a $(p \vee q) \implies r$ **b** Conditional

15

a $(p \implies q) \iff \sim r$ **b** Biconditional

16

a $(p \wedge q) \implies r$ **b** Conditional

17

a $(p \iff q) \vee r$ **b** Disjunction

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Additional questions

18 A biconditional statement is a statement that combines a conditional statement with its converse. This type of statement is constructed using the phrase "if and only if" which can be used to recognise it.

19 The following statements are biconditional

a A triangle is equilateral if and only if it is equiangular.

c It is Cinco de Mayo if and only if it is the 5th of May.

20 If a ray is an angle bisector then it divides an angle into two congruent angles.
If a ray divides an angle into two congruent angles then it is an angle bisector.

21 Two sides of a triangle are congruent if and only if the two angles opposite them are congruent.

22 A biconditional statement is true when the two implied conditonal statements are true.

23

a True. We know this because the two implied conditional statements:

- If a number is even, then it is divisible by two.
- If a number is divisible by two, then it is even.

are true.

b False. This is because the implied conditional statement:

- A number is greater than four if it is greater than three.

is false, with a counterexample being the number $3\frac{1}{2}$.

c True. We know this because the two implied conditional statements:

- If a number is prime, then it has no factors other than itself and one.
- If a number has no factors other than itself and one, then it is prime.

are true.

d False. This is because the implied conditional statement:

- If all the sides of a shape are congruent, then it is a regular polygon.

is false, with a counterexample being a rhombus.

24 If a quadrilateral is a parallelogram, then its diagonal bisect each other.

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If a quadrilateral has diagonals that bisect each other, then it is a parallelogram.

25

a A segment is a triangle median if and only if its endpoints are a vertex and the midpoint of the opposite side of a triangle.

b An angle is a right angle if and only if it measures exactly 90°.

c A three-dimensional figure is a sphere if and only if it consists of all points equidistant from a point called the center.

26 For a biconditional statement to be false, at least one of the implied conditional statements must be false. Consider that:

a Neva can use an umbrella even if it is not raining.

b It is possible to see both the sun and the moon at the same time. For example, before and after a solar eclipse.

c Deloris can choose to walk to school without having to miss the bus.

d Ulrich could get items from the top shelf without being tall by using a stool or a grabbing tool to reach them.

27

a Not a valid definition, since it is not reversible. The statement "if the measure of an angle is greater than 90°, then it's obtuse" is not true, so we cannot write this definition as an "if and only if" statement.

b Valid

c Not a valid definition, since the phrase "next to each other" is not precise.

d Not a valid definition, since the terms "vector" and "dot product" are not commonly understood or defined in this course.

28 No. Knowing that a biconditional statement is not true doesn't tell us which of the two implied conditional statements is not true.

29 A valid definition can be written as a biconditional statement because it matches a specific set of properties to an object (or scenario) in an exclusive relationship which can be represented by the phrase "if and only if".

30 0.1

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